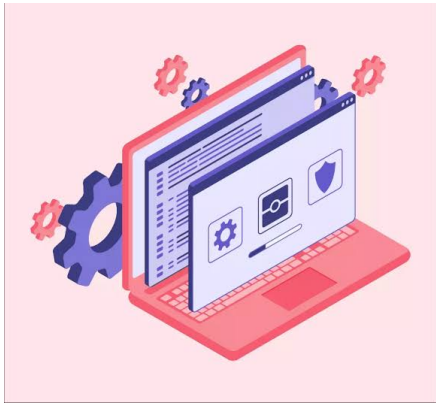
Operating System Structure & Virtual Machines



- System Structure
- Virtual Machines

Operating System Structure & Virtual Machines

Today's Topics

- Operating System Structure
- Monolithic Structure
- Layered Structure
- Microkernel
- Modular Structure
- Hybrid Structure
- Virtual Machines
- Hypervisors

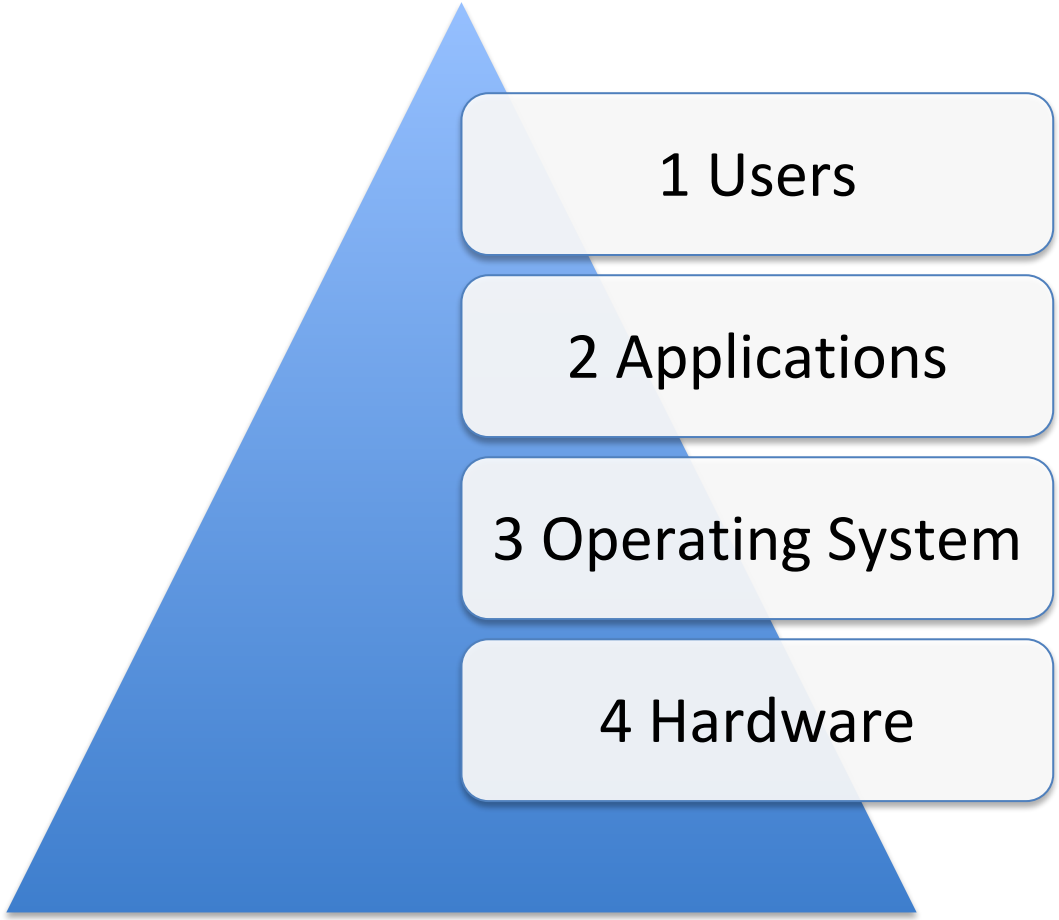
What is OS Structure?

Definition:
Operating System structure describes how the different components of an operating system are organized and interact with each other.

Why is structure important?
A good structure helps achieve:

- Modularity
- Maintainability
- Reliability
- Security
- Performance
- Easier debugging

Basic OS Architecture



Different OS Structures

Major Approaches

1. Simple Structure
2. Monolithic Structure
3. Layered Structure
4. Microkernel Structure
5. Modular Structure
6. Hybrid Structure

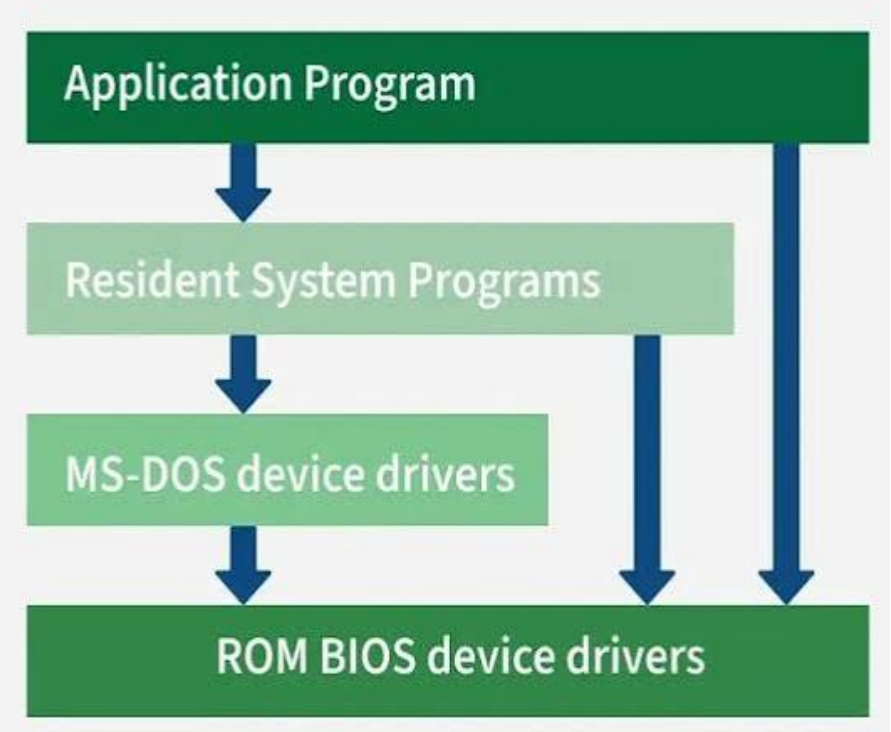
These structures represent different ways of organizing OS components.

1 Simple Structure

Concept
Early operating systems often had relatively little separation between components.

Characteristics
- Simple design
- Limited modularity
- Components can interact closely
- Difficult to maintain as the system grows

Example
MS-DOS



2 Monolithic Structure

Concept:
In a **monolithic kernel**, major OS services operate within a single large kernel address space.
Components:

Kernel
Process Management
Memory Management
File System
Device Drivers
Networking
I/O Management

Example:
Linux:
Linux uses a **monolithic kernel with a modular design**.

Monolithic Structure (Cont.)

Advantages

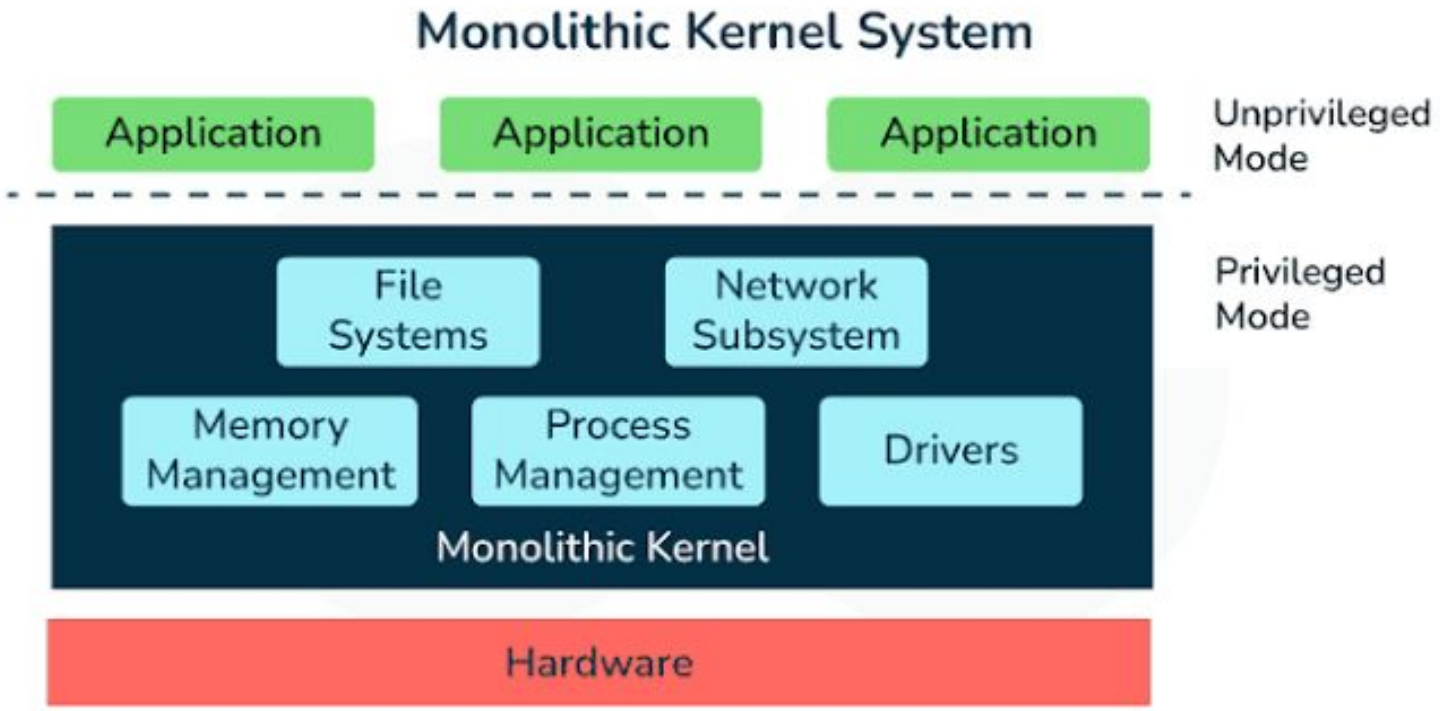
- High performance
- Fast communication between components
- Direct interaction between kernel components

Disadvantages

- Large kernel
- More difficult to maintain
- A kernel bug can potentially affect the entire system
- Less isolation between components

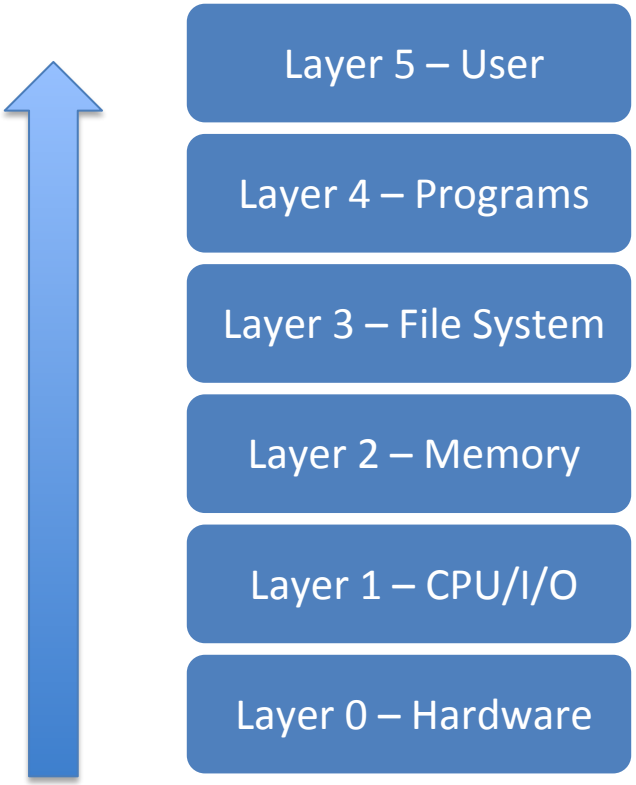
Key Idea

Most core OS services run in kernel space.


Monolithic Kernel System
Application
Application
Application
Unprivileged Mode
Privileged Mode
File Systems
Network Subsystem
Memory Management
Process Management
Drivers
Monolithic Kernel
Hardware

3 Layered Structure

Concept:
The Operating System is divided into layers.
Each layer uses services provided by the layer below it.



Principle:
Higher layer → uses lower layer

Layered Structure Example

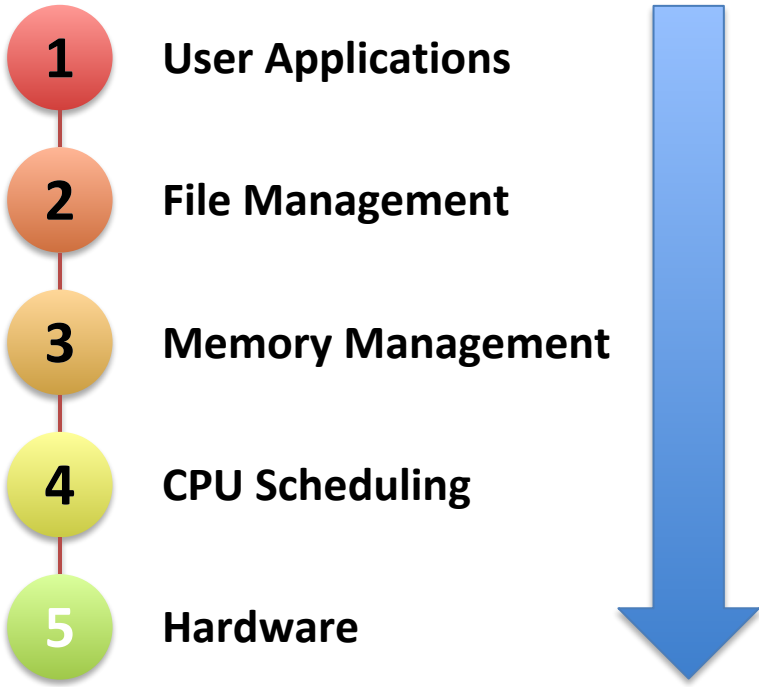
Imagine a simplified system:

Advantages

- Easier to understand
- Easier debugging
- Better modularity
- Easier maintenance

Disadvantages

- Difficult to define perfect layers
- Additional overhead may occur
- Some operations don't fit neatly into one layer



4 Microkernel Structure

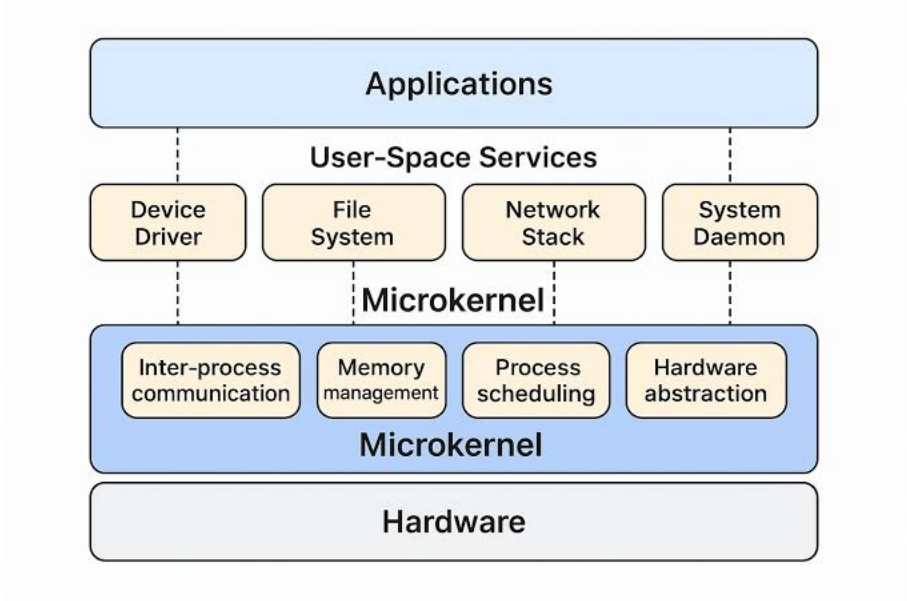
Concept:
A **microkernel** keeps only essential functionality inside the kernel.

Kernel may provide:

- Basic process management
- Basic memory management
- Inter-process communication
- Low-level hardware mechanisms

Other services can run outside the kernel.

Microkernel Architecture



Important Concept
Smaller kernel + services outside kernel

Advantages & Disadvantages of Microkernel

Advantages:

- Smaller kernel
- Better isolation
- Easier maintenance
- Improved fault isolation
- Better security potential
- Services can be developed independently

Disadvantage:

Communication overhead

Services may need to communicate with the microkernel and each other through IPC.

Microkernel Examples

Examples of systems using microkernel or microkernel-based designs include:

- QNX
- MINIX 3
- seL4

Important Clarification:
Microkernel designs can introduce IPC(Inter Process Communication) and context-switch overhead, although modern implementations can be highly optimized.

Monolithic vs Microkernel

FEATURE	MONOLITHIC	MICROKERNEL
KERNEL SIZE	Large	Small
CORE SERVICES	Mostly kernel space	Minimal kernel
PERFORMANCE	Generally high	IPC overhead may occur
ISOLATION	Lower	Higher
MAINTAINABILITY	More complex	Easier
EXAMPLE	Linux	QNX, MINIX 3

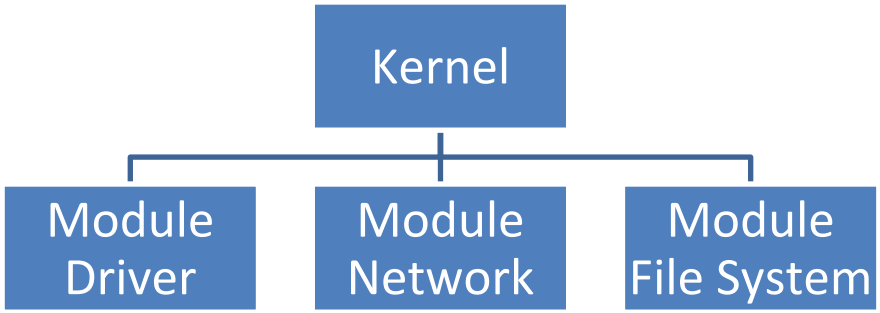

sebas@cupido:/home/sebas
$ ls -l /
total 36
drwxr-xr-x 2 bin operator 2880 May 3 2006 bin
drwxr-xr-x 4 root operator 448 May 3 2006 boot
drwxr-xr-x 2 root operator 17728 May 3 2006 dev
drwxr-xr-x 4 root operator 2112 Jul 2 20:50 etc
drwxrwxrwx 6 bin bin 384 Jul 2 19:39 home
drwxr-xr-x 3 root operator 192 May 3 2006 lib
drwxr-xr-x 3 256 operator 192 Jul 2 20:52 misc0
drwxrwxrwx 2 root operator 128 Oct 27 2006 mnt
drwx------ 3 root operator 1152 Jul 2 19:38 root
drwxr-xr-x 2 bin operator 640 May 3 2006 sbin
drwxrwxrwx 2 root operator 512 Jul 2 20:53 tmp
drwxr-xr-x 21 root operator 1344 Jun 30 12:46 usr
drwxr-xr-x 3 root operator 256 Oct 27 2006 var
$
Calculator
0
DEG
1/x x² √ CE/C AC
INV sin cos tan DRG
e EE log ln y^x
π x! ( ) ÷
STO 7 8 9 *
RCL 4 5 6 -
SUM 1 2 3 +
xclock
xv controls
UNREGISTERED
Display 24/8 Bit Algorithms
Root Windows Image Size
minix3.xpm
Next
Prev
Load
Save
Print
Delete
1 file
512x438 Xpm. 8-bit mode. Got all 150 colors.
Grab
Crop UnCrop AutoCrop About XV Quit
xv 3.10a: sebas/minix3.xpm <unregistered>
MINIX 3


Elektrobit
QNX
QNX Software Development Platform
EB corbos AdaptiveCore

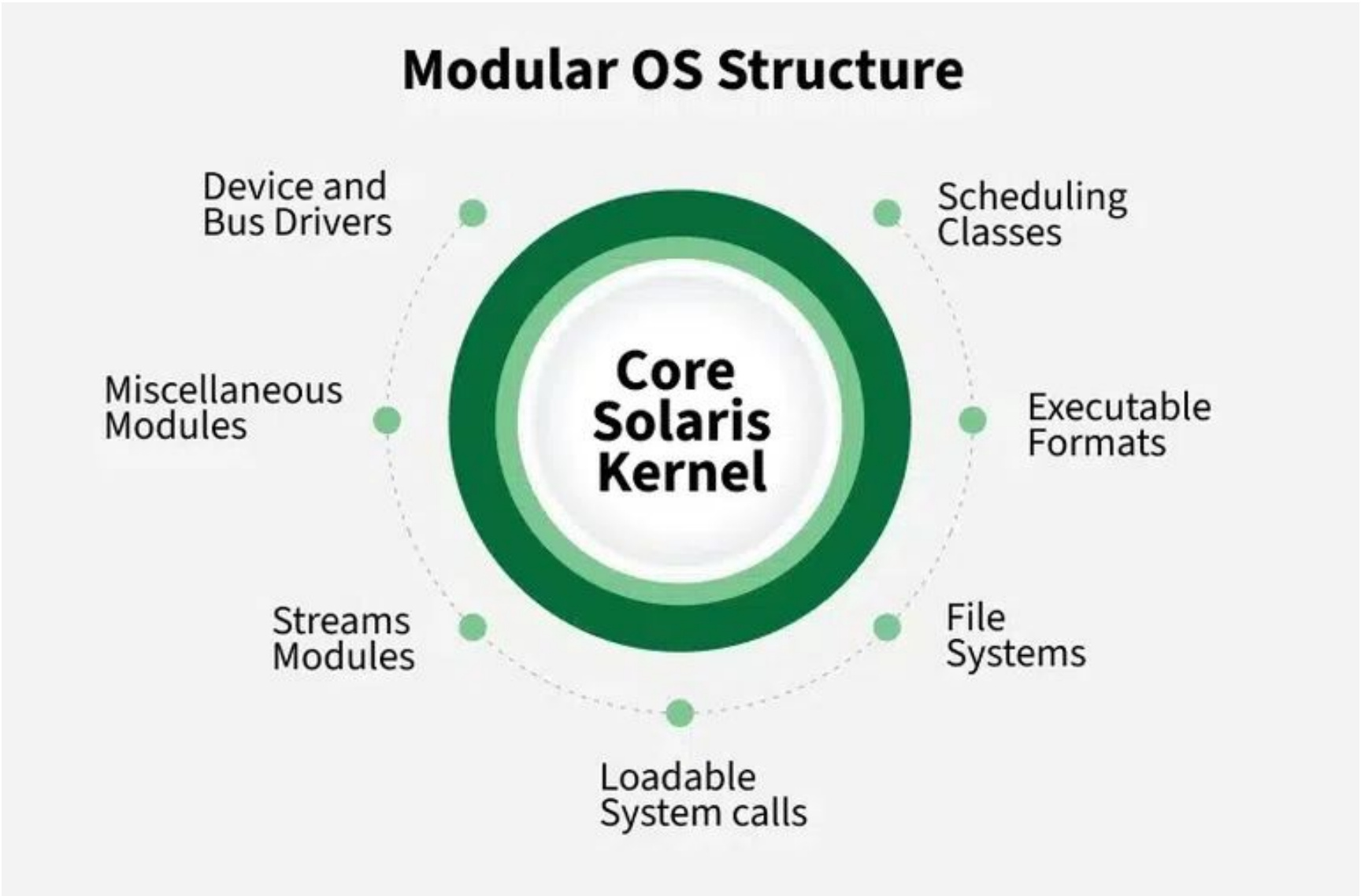
5 Modular Structure

Concept:
A modular kernel allows components to be added or removed dynamically.

Example:
Linux supports **Loadable Kernel Modules (LKMs)**.



Example:
A device driver can be loaded as a kernel module.


Modular OS Structure
Device and Bus Drivers
Scheduling Classes
Miscellaneous Modules
Core Solaris Kernel
Executable Formats
Streams Modules
File Systems
Loadable System calls

Advantages of Modular Structure

Advantages:

- Flexible
- Easier to extend
- Easier maintenance
- Components can be loaded when required
- Avoids rebuilding the entire kernel for every addition

Example:

Connecting a new hardware device may require loading an appropriate driver module.

6 Hybrid Structure

Concept:
A **hybrid kernel** combines ideas from different kernel architectures.
It aims to balance:

- Performance
- Modularity
- Flexibility
- Reliability

Examples:

- Windows NT family
- Apple's XNU

Important:
Hybrid designs don't mean "half monolithic + half microkernel" in a strict mathematical sense. They combine architectural ideas.

Hybrid Architecture



Applications

User Mode
handles other OS features

- Process Management
- File Servers
- I/O Management
- Memory Management

Kernel Mode
handles only basic OS features

- Device Drivers
- Low Level Address space Management
- Low Level process Management
- Inter process Communication

Hardware

OS Structure Comparison

STRUCTURE	MAIN IDEA	EXAMPLE
1 SIMPLE	Minimal organization	MS-DOS
2 MONOLITHIC	Large kernel	Linux
3 LAYERED	Hierarchical layers	THE system
4 MICROKERNEL	Minimal kernel	QNX
5 MODULAR	Loadable components	Linux
6 HYBRID	Combines approaches	Windows, XNU

❖ Virtual Machines

CAN ONE PHYSICAL COMPUTER RUN MULTIPLE OPERATING SYSTEMS AT THE SAME TIME?



What is a Virtual Machine?

Definition:
A **Virtual Machine (VM)** is a software-based representation of a physical computer that provides an environment in which an operating system and applications can run.

Example:

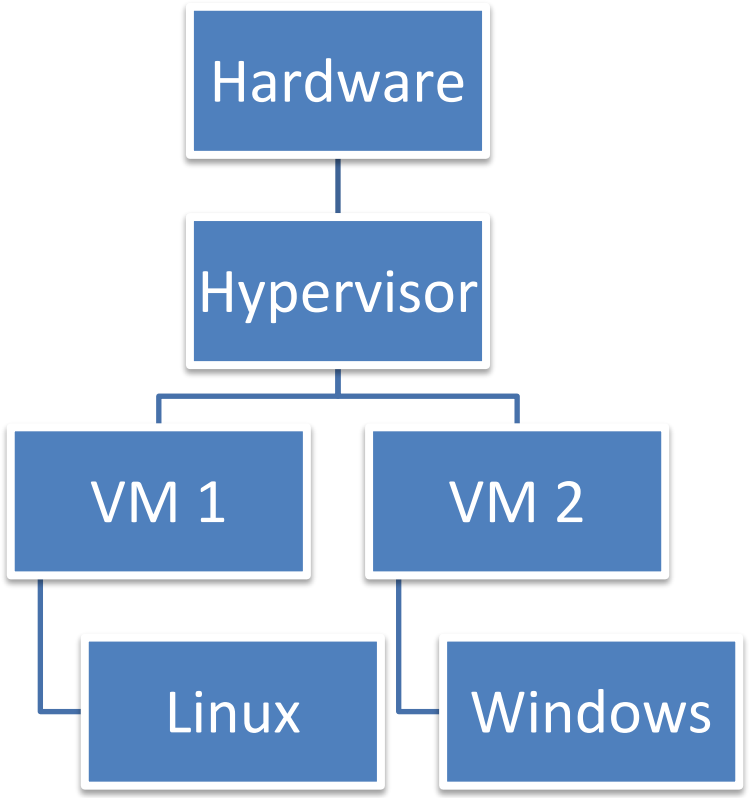
Physical Mac
↓
Virtual Machine
↓
Windows / Linux

Virtualization

Without Virtualization:



With Virtualization:

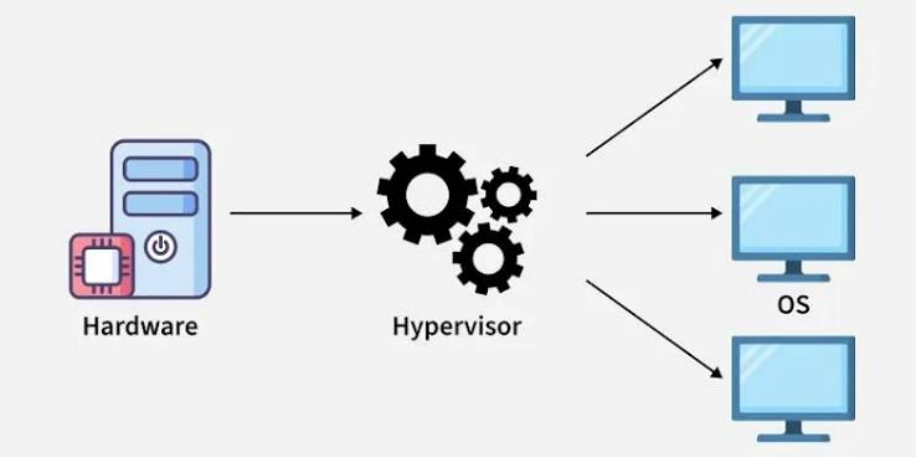


What is a Hypervisor?

Hypervisor:
A **hypervisor**, also called a **Virtual Machine Monitor (VMM)**, is software that creates and manages virtual machines.

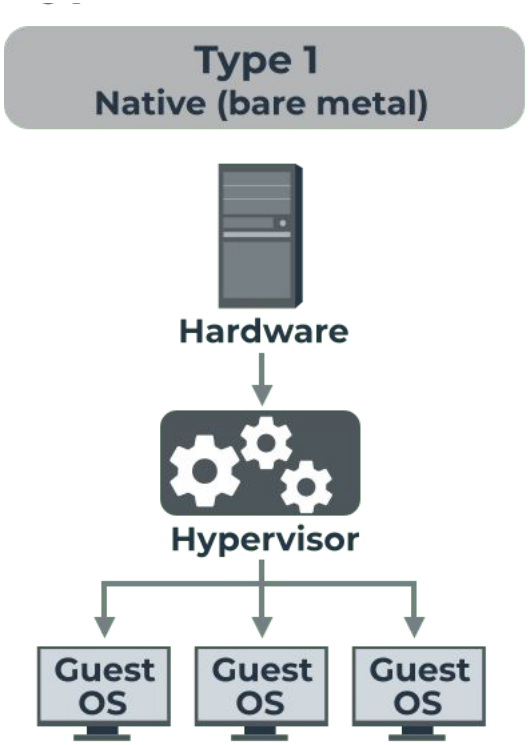
Responsibilities:

- Creates VMs
- Allocates CPU
- Allocates memory
- Manages virtual storage
- Manages virtual devices
- Isolates VMs



Type 1 Hypervisor

Bare-Metal Hypervisor:
Runs directly on physical hardware.



Examples:

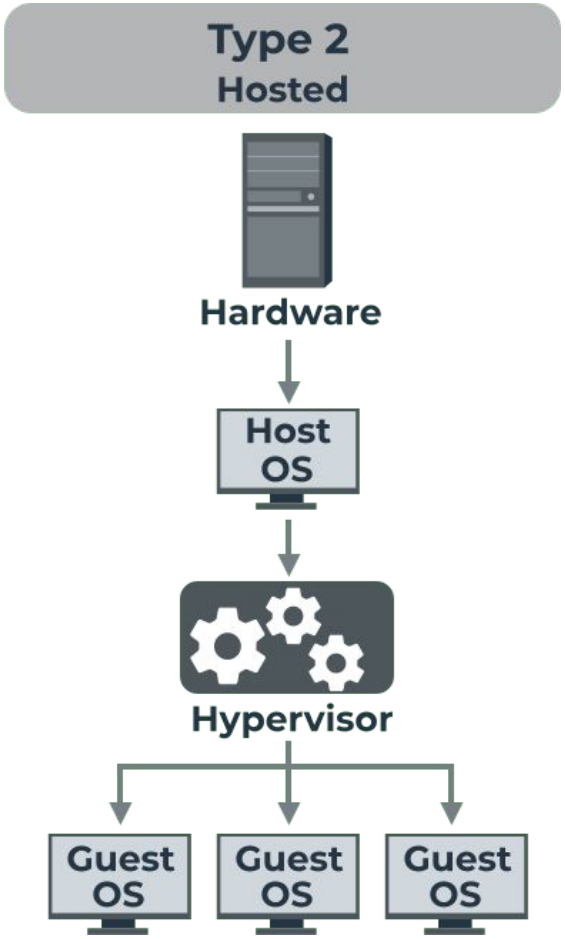
- VMware ESXi
- Microsoft Hyper-V
- Xen

Common Use:

- Data centers
- Cloud computing
- Enterprise servers

Type 2 Hypervisor

Hosted Hypervisor:
Runs as an application on a host operating system.



Examples

- VMware Fusion
- VMware Workstation
- Oracle VirtualBox

Common Use

- Personal computers
- Development
- Testing
- Education

Type 1 vs Type 2

FEATURE	TYPE 1	TYPE 2
1 Runs on	Hardware	Host OS
2 Also called	Bare-metal	Hosted
3 Typical use	Servers	Desktop
4 Performance	Generally higher	Additional host OS layer
5 Example	ESXi	VirtualBox

Type 1 → Hardware first
Type 2 → Host OS first

Why Use Virtual Machines?

Advantages:

- Run multiple OSs
- Isolation
- Testing
- Development
- Server consolidation
- Snapshots
- Easy experimentation

Example

A student can run:

Windows Host → Linux VM

without replacing Windows.

Disadvantages of Virtual Machines

Limitations:

- Performance overhead
- Requires additional RAM
- Requires additional storage
- CPU resources are shared
- Hardware access can be more complex

Example

Running a VM with 8 GB RAM requires that the host system has enough available memory.

Virtual Machine vs Physical Machine

FEATURE	PHYSICAL MACHINE	VIRTUAL MACHINE
1 Hardware	Physical	Virtualized
2 OS	Directly on hardware	Runs in virtual environment
3 Resources	Dedicated	Allocated virtually
4 Multiple OS	Usually one primary OS	Multiple VMs possible
5 Isolation	Physical	Software-based

Quick Quiz

Q1. Which OS structure keeps most services inside a large kernel?
A. Microkernel B. Monolithic C. Layered D. Distributed

Q2. Which structure keeps the kernel relatively small and moves many services outside it?
A. Monolithic B. Layered C. Microkernel D. Batch

Q3. Which hypervisor runs directly on physical hardware?
A. Type 1 B. Type 2 C. Type 3 D. Hosted

Q4. Which is an example of a Type 2 hypervisor?
A. VMware ESXi B. Xen C. VMware Fusion D. Bare-metal Hypervisor

Q5. What is the primary purpose of a hypervisor?
A. Compile programs B. Manage virtual machines C. Manage files D. Create system calls

B
C
A
C
B

Summary

Today We Learned

OS Structures

- Simple
- Monolithic
- Layered
- Microkernel
- Modular
- Hybrid

Virtualization

- Virtual Machine
- Hypervisor
- Type 1 Hypervisor
- Type 2 Hypervisor
- Advantages & limitations

Key Concept:

OS Structure = How OS components are organized

Virtualization = Creating virtual computing environments on physical hardware